

South African Statist. J. (1989) 23, 219-230

A CONFIDENCE INTERVAL FOR A MEASURE ASSOCIATED WITH THE COMPARISON OF A TREATMENT WITH A CONTROL

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Key words: Median test; survival time; coverage; simulation.

Summary: The problem of comparing a treatment with a control is considered. A measure that summarizes the differences between the treatment and the control, without making any parametric model assumptions about the underlying distributions, is proposed. The measure is a generalization of the one considered by Gart (1963). An asymptotic confidence interval for this measure is given and it is shown that the interval has a confidence coefficient approximately equal to $1 - \alpha$. The coverage of the proposed interval is examined in a simulation study.

1. INTRODUCTION

Researchers in various fields often conduct comparative experiments which provide data that are useful for comparing the effects of an experimental

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treatment with the effects of a standard or a control treatment. Part of the analysis of such data is directed towards the estimation of a "summary" measure of differences between the effects of the treatment and the control. For example, if survival time is the response variable on which the data are available, a measure of interest is the probability of survival under the experimental treatment beyond say, the median survival time under the control treatment. Such a measure is particularly useful when the data analysis is to be done without making any parametric model assumptions about the survival distributions. This paper deals with the estimation problem associated with such a measure.

Let F and G be the absolutely continuous distribution functions of the survival times under the control and the experimental treatment, respectively. Let p be a given positive fraction and let ξ_p be the p th quantile of F . In many applications p is taken to be $\frac{1}{2}$ so that $\xi_{\frac{1}{2}}$ is the median of F (called the control median) but we maintain full generality here. A measure of interest is

$$\theta_p = G(\xi_p).$$

Note that $1 - \theta_{\frac{1}{2}}$ is the probability of survival under the experimental treatment beyond the control median. If this probability is "sufficiently" greater than say, $\frac{1}{2}$, the experimenter may either recommend use of the experimental treatment or he may decide to investigate further about its beneficial effects. Thus, in general, estimation of θ_p is an important problem. In this paper interval estimation of θ_p is considered and a large sample confidence interval for θ_p is given. A confidence interval serves two important purposes in practice. First, it provides the experimenter with a range of values of θ_p compatible with the experiment and secondly, it allows the experimenter to perform a test of hypothesis, if so desired.

Gart (1963) considered interval estimation of θ_p and his paper is a motivation for the present work. Gart's large sample confidence interval is based on the median test statistic of Mathisen (1943). However, Gart inadvertently left out a factor in the large sample variance of the test statistic (see for example, Gastwirth, 1968; page 697). It will be seen that estimation

of this "omitted" factor poses additional problems in the derivation of the confidence interval.

2. NOTATION AND DEFINITIONS

Let $X_1 < X_2 < \dots < X_m$ be the ordered survival times in a random sample of size m from F and let $Y_1 < Y_2 < \dots < Y_n$ be the ordered survival times in a random sample of size n from G .

Let $\lim_{m,n \rightarrow \infty} n/m = \lambda$, where λ is a fixed quantity and let $\{r_m\}$ be a sequence of positive integers such that $\lim_{m \rightarrow \infty} (r_m/m) = p$; for example, $r_m/m \leq p < (r_m + 1)/m$. Let $U_{m,n}$ be the number of Y observations that do not exceed X_{r_m} and let $\{h_m\}$ be a sequence of positive real numbers such that

$$(1) \quad \lim_{m \rightarrow \infty} h_m = 0 \text{ and } \lim_{m \rightarrow \infty} m h_m^2 = \infty.$$

For example, one may take $h_m = m^{-\frac{1}{2}} h$, for some $h(> 0)$ free from m . Also, let $W_{m,n}$ be the number of Y values in $X_{r_m} \pm h_m$ and let f and g be the density functions corresponding to F and G , respectively.

3. CONFIDENCE INTERVAL FOR θ_p

First consider $U_{m,n}/n$ as an estimator of θ_p . In the appendix we prove that the estimator is consistent. Further, as in Gastwirth (1968) we can prove (a brief sketch of the proof is given in the appendix) the following result:

Theorem The random variable $V_{m,n} = n^{\frac{1}{2}}[n^{-1}U_{m,n} - \theta_p]$ is asymptotically normally distributed with mean zero and variance

$$(2) \quad \sigma^2 = \theta_p(1 - \theta_p) + \lambda p(1 - p)R_p,$$

where $R_p = g^2(\xi_p)/f^2(\xi_p)$.

It follows that a confidence interval for θ_p with approximate confidence coefficient $1 - \alpha$ is given by

$$(3) \quad |V_{m,n}/\sigma| \leq Z_{\alpha/2},$$

$Z_{\alpha/2}$ being the $(1 - \alpha/2)^{th}$ quantile of the standard normal distribution. The main difficulty in using (3) is that σ^2 depends on the unknown density functions. Incidentally, the factor R_p in σ^2 was omitted by Gart (1963). It is evident that without this factor one can find a confidence interval for θ_p from (3).

In order to use (3), a consistent estimator of the ratio R_p is needed. Hettmansperger and Malin (1975) considered an apparently similar problem relating to the consistent estimation of $g(\nu_p)/f(\xi_p)$, where ν_p is the p^{th} quantile of G . Their approach, however, can not be applied in the present context since the ratio under consideration involves $g(\xi_p)$ and not $g(\nu_p)$. In this paper a consistent estimator of R_p is obtained in two steps. First, results of Sen (1966) show that a consistent estimator of $1/f(\xi_p)$ is

$$S_m = m^{\frac{1}{2}}[X_{m-C_m+1} - X_{C_m}]/Z_{\alpha/2}'$$

where $[X_{C_m}, X_{m-C_m+1}]$ is a $100(1 - \alpha)\%$ confidence interval for ξ_p . Secondly, it can be shown that (details are given in the appendix) under some mild regularity conditions on F and G , a consistent estimator of $g(\xi_p)$ is

$$T_{m,n} = W_{m,n}/(2nh_m).$$

Hence, a consistent estimator of R_p is simply $(S_m T_{m,n})^2$. From Slutsky's theorem it follows that a confidence interval for θ_p with approximate confidence coefficient $1 - \alpha$ is given by

$$(4) \quad |V_{m,n}/\hat{\sigma}| \leq Z_{\alpha/2}'$$

where

$$\hat{\sigma}^2 = \theta_p[1 - \theta_p] + \lambda p(1 - p)(S_m T_{m,n})^2.$$

It is easy to see that the above confidence limits are given by the roots of the quadratic equation

$$a\theta_p^2 - b\theta_p + c = 0,$$

where $a = n + Z_{\alpha/2}^2$, $b = 2U + Z_{\alpha/2}^2$, $c = n^{-1}U^2 - Z_{\alpha/2}^2 \lambda p(1 - p)(S_m T_{m,n})^2$ and $U = U_{m,n}$.

An example is given in the next section to illustrate the calculation of the proposed confidence interval.

4. EXAMPLE

The following data from Bickel and Doksum (1977, page 390) are the blood cholesterol levels for men in two different age groups, 20-30 years and 40-50 years.

20-	135	222	251	260	269	235	386	252	352	173	156
30yrs:											
40-	294	311	286	264	277	336	208	346	239	172	254
50yrs:											

Let F and G correspond to the age groups 40-50 years and 20-30 years, respectively. We illustrate the calculation of the interval for $p = .5$, so that $\theta_{\frac{1}{2}}$ is the probability that the blood cholesterol level of a 20-30 year old man is less than the median blood cholesterol level of 40-50 year old men. For these data $m = n = 11$; we use $\alpha = .05$, $r_m = 6$ and $h = .25$.

First, we find $C_m = 1$ from the tables of Owen (1962). We also find $X_{r_m} = 277$, $U_{m,n} = 9$, $W_{m,n} = 0$, $S_m = 294.4351$ and $T_{m,n} = 0$. These yield $a = 14.8416$, $b = 21.8416$ and $c = 7.3636$. Thus, an approximately 95% confidence interval for $\theta_{\frac{1}{2}}$ is (.5204, .9513). It should be noted that with this interval one would reject the null hypothesis of the equality of F and G .

5. SIMULATION RESULTS

Several simulation experiments were conducted in order to examine the coverage of the proposed confidence interval for finite sample sizes. In this section we present the results of these simulations. The sample sizes used in these simulations were various combinations of 10, 15, 20, 30, 50, 100, 200. From practical considerations, we chose the median to be the control

quantile of interest, so that p was equal to .5.

In the first experiment F and G were exponential with means $1/\lambda_1$ and $1/\lambda_2$, respectively. Value of α was taken to be .05 and C_m values were obtained from the tables of Owen (1962). The simulations consisted of generating two samples for a given combination of values of $m, n, h, \lambda_1, \lambda_2$ and computing the confidence interval. This was repeated 5000 times and the proportion of times the proposed interval contained the true parameter was recorded. This proportion provides an empirical estimate of the nominal coverage probability .95. Various values of h were tried; $h = \frac{1}{4}$ was found to work the best. The results of these simulations are given in Tables I and II.

TABLE I
Empirical coverage for exponential
distributions with
 $\lambda_1 = \lambda_2 = 1, \alpha = .05, h = .25$ and $p = .5$

n	m						
	10	15	20	30	50	100	200
10	.947	.963	.966	.969	.973	.980	.978
15	.930	.948	.942	.951	.964	.961	.963
20	.935	.942	.941	.949	.957	.958	.954
30	.940	.943	.935	.945	.951	.952	.957
50	.940	.944	.938	.934	.942	.946	.958
100	.949	.953	.939	.942	.943	.946	.958
200	.957	.960	.943	.945	.952	.947	.951

In the second experiment samples were generated from the Weibull distributions with scale parameters λ_1, λ_2 and shape parameters ϕ_1 and ϕ_2 , respectively. As before, the proportion of times the proposed interval contained the true parameter value was recorded for various combinations of sample sizes. The results are given in Tables III and IV.

TABLE II
Empirical coverage for exponential
distributions with
 $\lambda_1 = 2, \lambda_2 = 1, \alpha = .05, h = .25$ and $p = .5$

<i>n</i>	<i>m</i>						
	10	15	20	30	50	100	200
10	.949	.958	.950	.960	.949	.946	.936
15	.954	.961	.952	.962	.968	.963	.955
20	.958	.957	.958	.960	.961	.961	.963
30	.962	.963	.954	.955	.958	.952	.959
50	.966	.962	.955	.951	.959	.959	.952
100	.975	.971	.961	.959	.958	.959	.961
200	.977	.973	.961	.960	.963	.955	.959

TABLE III
Empirical coverage for Weibull*
distributions with
 $\lambda_1 = \lambda_2 = 1, \phi_1 = \phi_2 = 2, \alpha = .05, h = .25$ and $p = .5$

<i>n</i>	<i>m</i>						
	10	15	20	30	50	100	200
10	.961	.964	.964	.969	.969	.974	.975
15	.946	.952	.947	.952	.964	.961	.959
20	.950	.953	.943	.945	.956	.958	.952
30	.957	.954	.941	.951	.957	.957	.949
50	.961	.963	.950	.944	.954	.952	.954
100	.970	.965	.953	.947	.957	.952	.954
200	.973	.968	.951	.953	.961	.958	.952

$$*F(x) = 1 - \exp\{-(\lambda_1 x)^{\phi_1}\}; G(y) = 1 - \exp\{-(\lambda_2 y)^{\phi_2}\}.$$

We conclude that the coverage of the proposed confidence interval is close to the nominal value for moderately large sample sizes (standard error with 5000 simulations at $\alpha = .05$ is .00308) and this is true for both the

exponential and the Weibull distributions. Further simulations for other distributions such as the Gamma yield basically same results.

TABLE IV
Empirical coverage for Weibull
distributions with

$\lambda_1 = 2, \phi_1 = 2, \lambda_2 = 1, \phi_2 = 1, \alpha = .05, h = .25$ and $p = .5$

n	m						
	10	15	20	30	50	100	200
10	.975	.979	.965	.964	.962	.956	.948
15	.977	.980	.975	.975	.973	.974	.979
20	.983	.985	.972	.972	.968	.970	.970
30	.982	.973	.967	.966	.966	.963	.957
50	.984	.972	.962	.961	.960	.963	.963
100	.992	.974	.971	.962	.963	.954	.958
200	.991	.976	.975	.965	.967	.959	.953

ACKNOWLEDGEMENTS

The authors would like to thank Professor M. M. Desu of the State University of New York at Buffalo for helpful discussions. Thanks are also due to the anonymous referees for their valuable comments and suggestions. The research of the first author was supported in part by a grant (number 1470) from the University of Alabama and also in part by a fellowship from the College of Commerce and Business Administration, University of Alabama.

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APPENDIX

We first prove the following result

Theorem 1. The estimator $n^{-1}U_{m,n}$ is a consistent estimator of θ_p .

Proof For notational simplicity, write $U_1 = n^{-1}U_{m,n}$, where it is understood that U_1 depends on both m and n . Note the given X_{r_m} , $U_{m,n}$ has a binomial distribution with parameters n and $G(X_{r_m})$. Hence

$$E[U_1|X_{r_m}] = G(X_{r_m}) \text{ and } V[U_1|X_{r_m}] = n^{-1}G(X_{r_m})\{1 - G(X_{r_m})\}.$$

One can write

$$E[U_1] = \int_0^1 \gamma(u) \phi_m(u) du,$$

where $\gamma(u) = GF^{-1}(u)$ and ϕ_m is the density function of the Beta $(r_m, m - r_m + 1)$ distribution. Let Z_m be the random variable whose density function is ϕ_m . Using the fact that $r_m/m \rightarrow p$ as $m \rightarrow \infty$, one can show that for every non-negative integer k , $E(Z_m^k) \rightarrow p^k$, as $m \rightarrow \infty$. Hence as $m \rightarrow \infty$, $Z_m \xrightarrow{D} p$ (cf. Rao, 1973; pp. 121). Since $\gamma(\cdot)$ is bounded, by Helly-Bray theorem (cf. Rao, 1973; pp. 117) it follows that $E(U_1) \rightarrow \theta_p$, as $m \rightarrow \infty$. One can also write

$$V(U_1) = n^{-1} \int_0^1 \gamma(u) \{1 - \gamma(u)\} \phi_m(u) du + \int_0^1 \gamma^2(u) \phi_m(u) du - \left[\int_0^1 \gamma(u) \phi_m(u) du \right]^2.$$

Using the fact that $Z_m \xrightarrow{D} p$, as $m \rightarrow \infty$, and applying Helly-Bray theorem, it is easy to see that as $m, n \rightarrow \infty$, $V(U_1) \rightarrow 0$. Hence the result.

We now give a sketch of the proof of the asymptotic normality of $V_{m,n}$.

Theorem 2. The random variable $V_{m,n}$ is asymptotically normally distributed with mean 0 and variance σ^2 , where σ^2 is defined in (2).

Proof Define the following quantities

$$e_X = \sum_{i=1}^m \{I(X_i \leq \xi_p) - p\} / \{mp(1-p)\}^{\frac{1}{2}} \\ \text{and } e_Y = \sum_{j=1}^n \{I(Y_j \leq \xi_p) - \theta_p\} / \{n\theta_p(1-\theta_p)\}^{\frac{1}{2}},$$

where I denotes the usual indicator function. Using arguments similar to those in Gastwirth (1968; Lemma 3.1) one can show that as $m, n \rightarrow \infty$, such that $n/m \rightarrow \lambda$,

$$X_{r_m} - \xi_p = [f(\xi_p) \{m/(p(1-p))\}^{\frac{1}{2}}]^{-1} e_X + o_p(m^{-\frac{1}{2}})$$

and $n^{-1}U_{m,n} - \theta_p$

$$= [\theta_p(1-\theta_p)n^{-1}]^{\frac{1}{2}} e_Y + [R_p p(1-p)m^{-1}]^{\frac{1}{2}} e_X + o_p(m^{-\frac{1}{2}}).$$

From these it follows that as $m, n \rightarrow \infty$, such that $n/m \rightarrow \lambda$,

$$\begin{aligned} V_{m,n} &= n^{\frac{1}{2}}[n^{-1}U_{m,n} - \theta_p] \\ &= \{\theta_p(1 - \theta_p)\}^{\frac{1}{2}}e_Y + \{\lambda p(1 - p)R_p\}^{\frac{1}{2}}e_X + o_P(1). \end{aligned}$$

Thus $V_{m,n}$ is distributed approximately as a linear function of two independent random variables e_X and e_Y , each of which is distributed approximately as a standard normal variable. Proof of Theorem 2 follows.

Next we prove consistency of $T_{m,n}$.

Theorem 3. $T_{m,n} = W_{m,n}/(2nh_m)$ is a consistent estimator of $g(\xi_p)$.

Proof Assume that there exist a'_1, a'_2 ($a'_1 < \xi_p < a'_2$) such that (i) $G(y)$ admits a continuous derivative $g(y)$ for $y \in [a'_1, a'_2]$ and (ii) $F(x)$ is strictly increasing over $x \in [a'_1, a'_2]$. We need the following lemmas.

Lemma 1 Let $\beta_m(z) = (2h_m)^{-1}[G\{F^{-1}(z) + h_m\} - G\{F^{-1}(z) - h_m\}]$. Then $\beta_m(z) \rightarrow g(\xi_p)$, as $z \rightarrow p$ and $m \rightarrow \infty$.

Proof Let a_1, a_2 be such that $a'_1 < a_1 < \xi_p < a_2 < a'_2$ and $p_i = F(a_i)$, $i = 1, 2$. Note that by (i) above, $\lim_{m \rightarrow \infty} [(2h_m)^{-1}\{G(y+h_m) - G(y-h_m)\}] = g(y)$, uniformly in $y \in [a_1, a_2]$ and that by (ii) above $F^{-1}(z) \in [a_1, a_2]$, whenever $z \in [p_1, p_2]$. Hence $\lim_{m \rightarrow \infty} \beta_m(z) = g\{F^{-1}(z)\}$ uniformly in $z \in [p_1, p_2]$. Further, by (ii), $F^{-1}(z)$ is continuous at $z = p$ so that by (i), $g\{F^{-1}(z)\} \rightarrow g(\xi_p)$ as $z \rightarrow p$. The lemma follows since $p \in [p_1, p_2]$ by (ii), and $|\beta_m(z) - g(\xi_p)| \leq |\beta_m(z) - g\{F^{-1}(p)\}| + |g\{F^{-1}(z)\} - g(\xi_p)|$.

Lemma 2 For every $\delta > 0$ and $j = 0, 1, 2$,

$$h_m^{-j} P[|Z_m - p| \geq \delta] \rightarrow 0, \text{ as } m \rightarrow \infty,$$

where Z_m is defined as in the proof of Theorem 1.

Proof Take arbitrary $\delta > 0$. Since $E(Z_m) = r_m/(m+1)$ and $\lim(r_m/m) = p$, there exists m^* such that $|E(Z_m) - p| \leq \delta/2$, whenever $m \geq m^*$.

Hence for $m \geq m^*$, $|Z_m - p| \geq \delta \implies |Z_m - E(Z_m)| \geq \delta/2$. Since $\text{Var}(Z_m) = r_m(m - r_m + 1)/\{(m+1)^2(m+2)\}$ and $\lim_{m \rightarrow \infty} (r_m/m) = p$, the lemma follows from (1) and Chebyshev's inequality.

We will now prove the theorem by showing that as $m, n \rightarrow \infty$, $E(T_{m,n}) \rightarrow g(\xi_p)$ and $\text{Var}(T_{m,n}) \rightarrow 0$. By conditioning on X_{r_m} , it can be seen that

$$(A.1) \quad E(T_{m,n}) - g(\xi_p) = \int_0^1 \{\beta_m(z) - g(\xi_p)\} \phi_m(z) dz.$$

Since G is a cumulative distribution function and hence bounded by 0 and 1, one obtains $0 \leq \beta_m(z) \leq h_m^{-1}$, so that

$$(A.2) \quad |\beta_m(z) - g(\xi_p)| \leq h_m^{-1} + g(\xi_p), \text{ for each } z.$$

Consider an arbitrary $\epsilon > 0$. By Lemma 1, there exist $\delta(> 0)$ and m_0 , depending on ϵ , such that

$$(A.3) \quad |z - p| < \delta, m \geq m_0 \implies |\beta_m(z) - g(\xi_p)| < \epsilon.$$

By (A.1)-(A.3) we have for $m \geq m_0$,

$$\begin{aligned} |E(T_{m,n}) - g(\xi_p)| &\leq \int_{|z-p|<\delta} |\beta_m(z) - g(\xi_p)| \phi_m(z) dz \\ &\quad + \int_{|z-p|\geq\delta} |\beta_m(z) - g(\xi_p)| \phi_m(z) dz. \\ &< \epsilon + \{h_m^{-1} + g(\xi_p)\} P[|Z_m - p| \geq \delta]. \end{aligned}$$

By Lemma 2 the right hand side of the above inequality tends to ϵ , as $m \rightarrow \infty$. Since ϵ is arbitrary, it follows that $E(T_{m,n}) \rightarrow g(\xi_p)$ as $m \rightarrow \infty$.

By conditioning on X_{r_m} , next observe that $\text{Var}(T_{m,n}) = R_{2m} - R_{1m}^2 + R_{3m}$, where $R_{im} = \int_0^1 \{\beta_m(z)\}^i \phi_m(z) dz$, $i = 1, 2$ and $R_{3m} = n^{-1} \int_0^1 \beta_m(z) \{(2h_m)^{-1} - \beta_m(z)\} \phi_m(z) dz$. Exactly as before, using Lemmas 1 and 2, both R_{2m} and R_{1m}^2 tend to $\{g(\xi_p)\}^2$ as $m \rightarrow \infty$. Also, since $0 \leq \beta_m(z) \leq h_m^{-1}$ for each z , and n/m tends to λ as $n, m \rightarrow \infty$, it follows from (1) that

$$\begin{aligned} |R_{3m}| &\leq n^{-1} h_m^{-1} \{(2h_m)^{-1} + h_m^{-1}\} \\ &= (3/2)(nm^{-1})^{-1} (mh_m^2)^{-1} \rightarrow 0, \text{ as } m, n \rightarrow \infty. \end{aligned}$$

Hence $\text{Var}(T_{m,n}) \rightarrow 0$ as $m, n \rightarrow \infty$, completing the proof.